

[Mr Chu's
Solutions] 4052
2025 P1

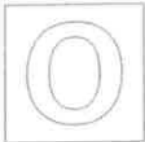
Thursday, October 23, 2025 9:01 AM



2025 GCE O
level EM P1

* 0022221558301 *





MINISTRY OF EDUCATION, SINGAPORE
in collaboration with
CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION
General Certificate of Education Ordinary Level

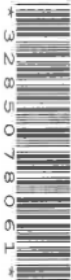
CANDIDATE NAME



CENTRE NUMBER

S

INDEX NUMBER



3
2
8
5
0
7
8
0
6
1

MATHEMATICS

Paper 1

4052/01

October/November 2025

2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE ON ANY BARCODES.

Answer **all** the questions.
The number of marks is given in brackets [] at the end of each question or part question.

If working is needed for any question it must be shown with the answer.
Omission of essential working will result in loss of marks.
The total of the marks for this paper is 90.

The use of an approved scientific calculator is expected, where appropriate.
If the degree of accuracy is not specified in the question and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.
For π , use either your calculator value or 3.142.

This document consists of 19 printed pages and 1 blank page.



Singapore Examinations and Assessment Board

© UCLES & MOE 2025



Cambridge Assessment
International Education

DC (PQ/CT) 343365/7

[Turn over]

* 0022221558302 *



2

Mathematical Formulae

Compound interest

$$\text{Total amount} = P \left(1 + \frac{r}{100} \right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi r l$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3}\pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3}\pi r^3$$

$$\text{Area of triangle } ABC = \frac{1}{2}ab \sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2}r^2\theta, \text{ where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

$$\overline{\sin A} = \overline{\sin B} = \overline{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Statistics

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

$$\text{Standard deviation} = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$

© UCLES & MOE 2025



4052/01/O/N/25

DO NOT WRITE IN THIS MARGIN

* 0022221558303 *



3

Answer **all** the questions.

1 $W = \frac{\sqrt{a}}{b+c}$

Calculate the value of W when $a = 1234$, $b = 5.67$ and $c = 8.9$.

Answer **2.41** [1]

2 These ingredients are needed to make 8 scones.

275 g flour
60 g butter
150 ml milk
30 g sugar

(a) Cheng uses 150 g of butter to make some scones.

How many scones does he make?

$$\frac{150}{60} \times 8 = 20$$

Answer **20** [1]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

Answer 20 [1]

(b) Mei has 700 ml of milk and 900 g of each of the other ingredients.

(i) Calculate the maximum number of scones she can make.

$$\frac{700}{18.75} = 37.3$$

$$\frac{900}{34.375} = 26.2$$

1 scone requires
 $275 \div 8 = 34.375$ g of flour
 $150 \div 8 = 18.75$ ml of milk

Answer 26 [2]

(ii) Calculate the amount of butter she uses to make this maximum number of scones.

$$\frac{26}{8} \times 60 = 195$$

Answer 195 g [1]



* 0022221558304 *



4

- 3 Samira invests \$5250 for 4 years.
 The money earns simple interest at a rate of 2.4% per year.

Find the total amount of money she has at the end of 4 years.

$$\frac{100 + (2.4 \times 4)}{100} \times \$5250 = \$6930$$

Answer \$ 6930 [2]

- 4 Chen is selling 19.2 grams of gold.
 He is offered \$58.11 per gram.

Each of these calculations can be used to estimate how much money he will receive.

A. 58.11 B. 60.00 C. 55.00

Each of these calculations can be used to estimate how much money he will receive.

A: 58×19

B: 60×20

C: 55×20

Which of these calculations overestimates the amount he receives?
Without doing any calculation, explain how you know.

Calculation B because 19.2 is rounded up to 20 and 58.11 is rounded up to 60. [1]



* 0022221558305 *



5

5 (a) Solve $3c - 4 = 9c + 17$.

$$\begin{aligned} -6c &= 21 \\ c &= 21 \div -6 \\ &= -3.5 \end{aligned}$$

Answer $c =$ -3.5 [2]

(b) Factorise $8ab - 3 + 12a - 2b$.

Answer $c = \dots -5.5 \dots$ [2](b) Factorise $8ab - 3 + 12a - 2b$.

$$\begin{aligned}
 &= 8ab + 12a - 3 - 2b \\
 &= 4a(2b+3) - 1(2b+3) \\
 &= (4a-1)(2b+3)
 \end{aligned}$$

Answer $(4a-1)(2b+3)$ [2](c) Expand and simplify $(4x+3y)(3x-4y)$.

$$\begin{aligned}
 &= 12x^2 + 9xy - 12xy - 12y^2 \\
 &= 12x^2 - 3xy - 12y^2
 \end{aligned}$$

Answer $12x^2 - 3xy - 12y^2$ [2]6 (a) Solve the inequalities $-43 < 2(5x+3) - 14 \leq 42$.

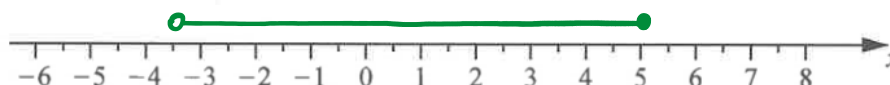
$$-43 < 10x + 6 - 14 \leq 42$$

- 6 (a) Solve the inequalities $-43 < 2(5x+3) - 14 \leq 42$.

$$\begin{aligned}
 -43 &< 10x+6-14 \leq 42 \\
 -43 &< 10x-8 \leq 42 \\
 +8 &\downarrow \\
 -35 &< 10x \leq 50 \\
 \div 10 &\downarrow \\
 -3.5 &< x \leq 5
 \end{aligned}$$

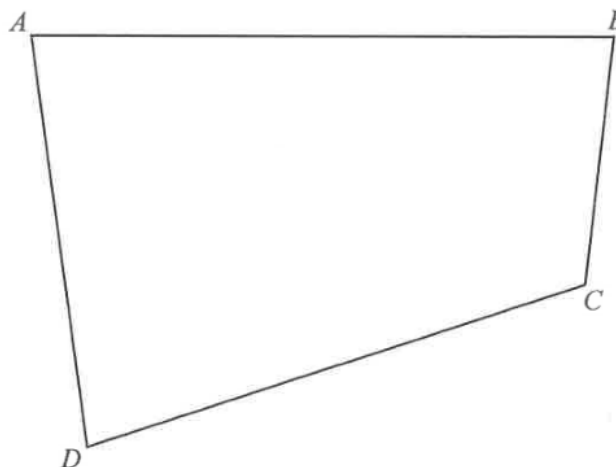
Answer $-3.5 < x \leq 5$ [3]

- (b) Represent your answer to part (a) on this number line.



[1]

- 7 The diagram shows a quadrilateral $ABCD$.



- (a) Construct the perpendicular bisector of AB . [1]
- (b) Construct the bisector of angle ADC . [1]
- (c) A point P is to be plotted inside the quadrilateral.
It is to be closer to A than B , and closer to CD than AD .

Shade the region where P can be plotted.

[1]





- 8 P is the point $(4, -1)$.

$$\overrightarrow{PQ} = \begin{pmatrix} -7 \\ 10 \end{pmatrix}.$$

- (a) Find the coordinates of point Q .

$$\begin{aligned} \vec{OQ} &= \vec{OP} + \vec{PQ} \\ &= \begin{pmatrix} 4 \\ -1 \end{pmatrix} + \begin{pmatrix} -7 \\ 10 \end{pmatrix} = \begin{pmatrix} -3 \\ 9 \end{pmatrix} \end{aligned}$$

Answer (..... -3 9) [1]

- (b) Find the magnitude of $\overrightarrow{PQ}$.

$$|\overrightarrow{PQ}| = \sqrt{(-7)^2 + 10^2} = 12.2 \text{ units (3 s.f.)}$$

Answer 12.2 units [2]

- 9 The total surface area of a cube is 125 cm^2 .

Calculate the volume of the cube.

$$\begin{aligned} \text{volume} &= \left(\sqrt{\frac{125}{6}} \right)^3 \\ &= 95.1 \text{ cm}^3 \text{ (3 s.f.)} \end{aligned}$$

Answer 95.1 cm^3 [2]

- 10 (a) Written as a product of its prime factors, $5500 = 2^x \times 5^y \times 11$.

Find the values of x and y .

$$\begin{array}{r|l} 11 & 5500 \\ \hline 2 & 500 \\ \hline 2 & 250 \\ \hline 5 & 125 \\ \hline 5 & 25 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

Answer $x = \dots\dots\dots 2 \dots\dots\dots$, $y = \dots\dots\dots 3 \dots\dots\dots$ [1]

- (b) Written as a product of its prime factors, $793\,800 = 2^3 \times 3^4 \times 5^2 \times 7^2$.

Find the smallest positive integer k such that $793\,800k$ is a cube number.

$$\begin{aligned} k &= 3^2 \times 5 \times 7 \\ &= 315 \end{aligned}$$

Answer $k = \dots\dots\dots 315 \dots\dots\dots$ [1]

$$= 315$$

Answer $k = 315$ [1]

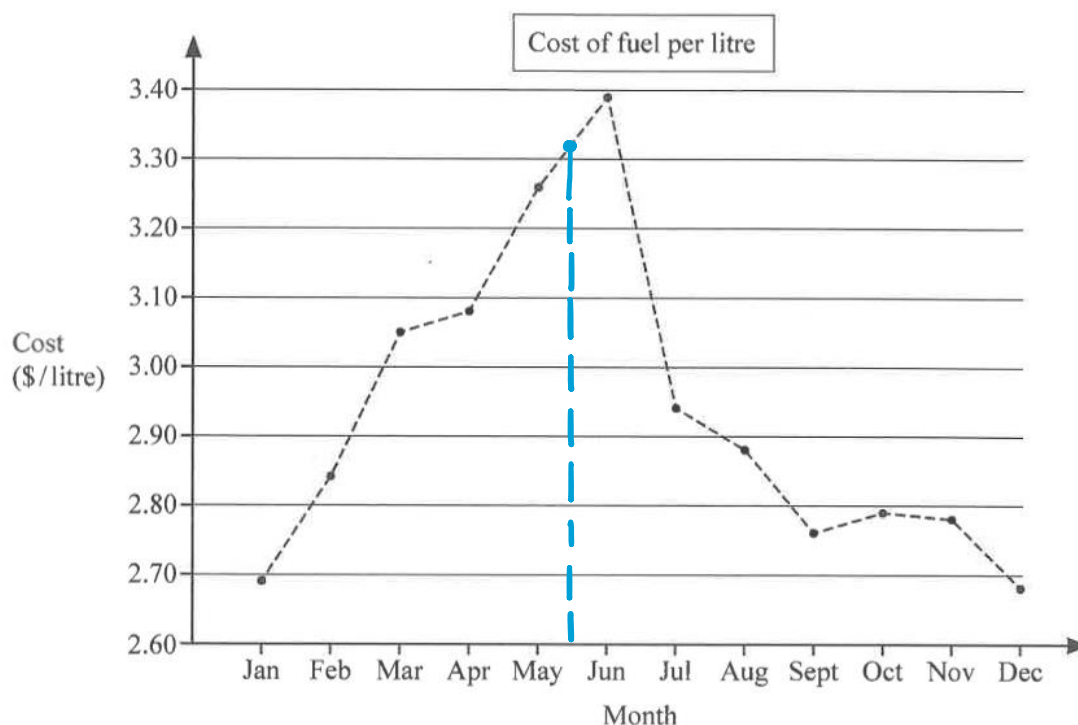


* 0022221558308 *



8

- 11 Alex records the cost of fuel per litre at the end of each month. He draws this graph to show the information.



- (a) Sue says that the cost of fuel per litre in the middle of June was about \$3.32.

Explain why she may be wrong.

While the line graph shows a value of \$3.32 between end of May and End of June, Sue may be wrong as the line graph data is assumed between end of May and end of June. [1]

- (b) Explain why the graph may not show the highest cost per litre.

The line graph only shows data at the end of each month and the highest cost per litre may have occurred at the middle of any month. [1]

- (c) (i) State one misleading feature of the graph.

The vertical axis does not start from 0. [1]

(ii) Explain how this feature affects the reader's interpretation of the graph.

- (ii) Explain how this feature affects the reader's interpretation of the graph. [1]

The increase in cost per litre becomes exaggerated. For example, the fuel pile seems to have increased by 700% from ^{end of} January

to end of June when in actual fact it only increased by about $\frac{3.39 - 2.69}{2.69} \times 100\% \approx 26\%$. [1]

© UCLES & MOE 2025

4052/01/O/N/25

* 0022221558309 *



9

12

$$8a - 4b + 7 = 6c$$

Explain why a , b and c cannot all be integers.

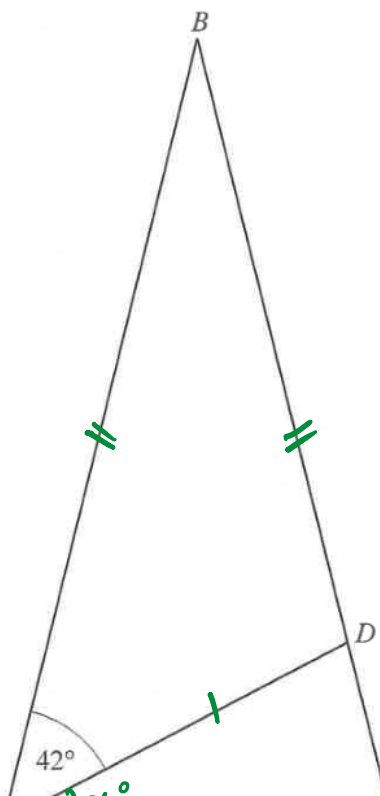
if a and b are integers, $8a - 4b$ must be even.

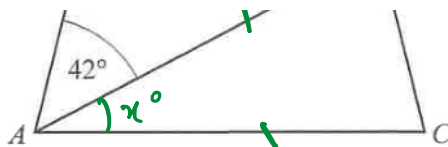
$8a - 4b + 7$ must be odd. However, $6c$ must be even if

c is an integer. Since it is not possible for $6c$ to be odd and even, [1]

a , b and c cannot all be integers.

13





In the diagram, ABC is an isosceles triangle with $AB = BC$.
 D is a point on BC such that $AD = AC$.
 Angle $BAD = 42^\circ$.

Find angle ACD .

Let $\angle DAC = x^\circ$.

$\angle DCA = (42 + x)^\circ$ (base angles of isos. $\triangle ABC$)

$\angle ADC = (42 + x)^\circ$ (base angles of isos. $\triangle ADC$)

$42 + x + 42 + x + x = 180$

$\angle ACD = 42 + 32 = 74^\circ$

$\Rightarrow x = 180 - 42 - 42$
 $= 96$

Answer Angle $ACD = 74^\circ$ [3]



* 0022221558310 *



10

- 14 From 2018 to 2021, Singapore's Gross National Income (GNI) increased by 9.24%.

From 2018 to 2019, the GNI increased by 3.63%.

From 2019 to 2020, the GNI decreased by 5.56%.

Calculate the percentage increase in the GNI from 2020 to 2021.

Let percentage increase from 2020 to 2021 be $x\%$.

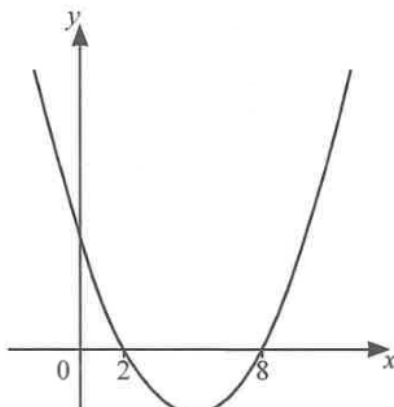
$$\frac{103.63}{100} \times \frac{100 - 5.56}{100} \times \frac{100 + x}{100} = \frac{109.24}{100}$$

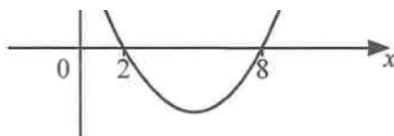
$$0.97868 (100 + x) = 109.24$$

$$x = \frac{109.24}{0.97868} - 100 = 11.6$$

Answer 11.6 % [3]

15





The diagram shows a sketch of a quadratic curve.

- (a) It is given that the equation for the curve is of the form $y = x^2 + bx + c$.

Find the value of b and the value of c .

$$(x-2)(x-8) = x^2 - 10x + 16$$

Answer $b = -10$, $c = 16$ [2]

- (b) A different quadratic curve also has roots $x = 2$ and $x = 8$.
The intercept on the y -axis is 32.

Find an equation for this curve.

Let equation be $y = a(x-2)(x-8)$
When $x = 0$, $y = 32$

$$32 = a(0-2)(0-8)$$

$$32 = 16a$$

$$a = 2$$

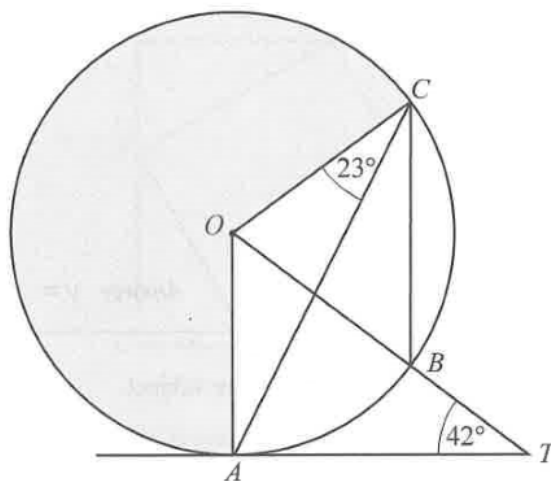
Answer $y = 2(x-2)(x-8)$
or $y = 2x^2 - 20x + 32$ [1]

* 0022221558311 *



11

16



A , B and C are points on the circumference of a circle, centre O .
 TA is a tangent to the circle.
 OBT is a straight line.
Angle $OTA = 42^\circ$ and angle $OCA = 23^\circ$.

(a) Give a reason why angle $OAT = 90^\circ$.

ODI is a straight line.

Angle $OTA = 42^\circ$ and angle $OCA = 23^\circ$.

- (a) Give a reason why angle $OAT = 90^\circ$.

OA , the radius, is perpendicular to the tangent AT at A .

[1]

- (b) Find angle ACB .

Give a reason for each step of your working.

$$\angle AOT = 180^\circ - 90^\circ - 42^\circ = 48^\circ \text{ (angle sum of } \triangle AOT, \text{ tangent perpendicular to radius)}$$

$$\angle ACB = 48^\circ \div 2 = 24^\circ$$

(angle at centre = $2 \times$ angle at circumference)

Answer Angle $ACB = 24^\circ$ [2]

- (c) The radius of the circle is 8.7 cm.

Calculate the area of the shaded sector OAC .

$$\angle COA = 180^\circ - 23^\circ \times 2 = 134^\circ$$

$$\begin{aligned} \text{Area} &= \frac{360^\circ - 134^\circ}{360^\circ} \times \pi (8.7)^2 \\ &= 149 \text{ cm}^2 \text{ (3 s.f.)} \end{aligned}$$

Answer 149 cm^2 [3]



* 0022221558312 *



12

- 17 Given that $343^4 = 49^{-5y}$, find y .

$$(7^3)^4 = (7^2)^{-5y}$$

$$7^{12} = 7^{-10y}$$

$$12 = -10y$$

$$y = -1.2$$

Answer $y = -1.2$ [2]

- 18 Rearrange the formula $\frac{a}{2a-3} = \frac{3c}{d} + 1$ to make a the subject.

- 18 Rearrange the formula $\frac{a}{2a-3} = \frac{3c}{d} + 1$ to make a the subject.

$$\frac{a}{2a-3} = \frac{3c+d}{d}$$

$$ad = (2a-3)(3c+d)$$

$$ad = 6ac - 9c + 2ad - 3d$$

$$ad - 6ac - 2ad = -9c - 3d$$

$$-6ac - ad = -9c - 3d$$

$$\times -1: \quad 6ac + ad = 9c + 3d$$

$$a(6c+d) = 9c+3d$$

$$a = \frac{9c+3d}{6c+d}$$

Answer $a = \frac{9c+3d}{6c+d}$ [3]

- 19 By completing the square, find the coordinates of the minimum point of the curve $y = x^2 + 6x + 11$. Show all your working.

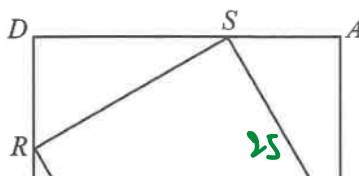
$$y = x^2 + 6x + 11$$

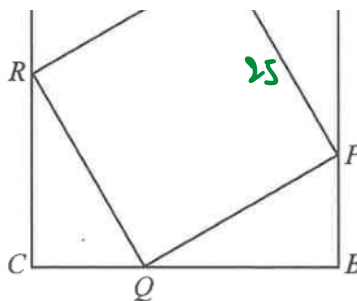
$$= (x+3)^2 - 3^2 + 11$$

$$= (x+3)^2 + 2$$

$$\text{coordinates of min. point} = (-3, 2)$$

Answer (-3 , 2) [3]





The diagram shows two squares, $ABCD$ and $PQRS$.

- (a) Show that triangle APS and triangle BQP are congruent.
Give a reason for each statement you make.

Answer

$$\angle SAP = \angle PBQ = 90^\circ \text{ (int. angle of square} = 90^\circ)$$

$$\angle APS = 180^\circ - 90^\circ - \angle QPB$$

$$= 180^\circ - 90^\circ - (180^\circ - 90^\circ - \angle BQP)$$

$$= \angle BQP$$

$$PS = QP \text{ (sides of square are equal)}$$

$\therefore \triangle APS$ and $\triangle BQP$ congruent (AAS congruency test)

[3]

- (b) The perimeter of the larger square is 124 cm.
The perimeter of the smaller square is 100 cm.

Find the length of AP and the length of PB .

$$AB = 124 \div 4 = 31 \text{ cm.}$$

$$\text{Let } AP = x \text{ cm. } \therefore PB = (31 - x) \text{ cm.}$$

$$AS = (31 - x) \text{ cm.}$$

using right-angled $\triangle APS$,

$$25^2 = x^2 + (31 - x)^2$$

$$625 = x^2 + 961 - 62x + x^2$$

$$0 = 2x^2 - 62x + 336$$

$$0 = x^2 - 31x + 168$$

$$(x - 24)(x - 7) = 0$$

$$x = 24 \text{ or } x = 7 \text{ (rej.)}$$

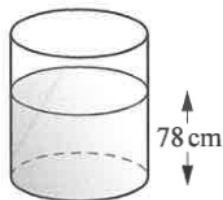
$$PB = 31 - 24 = 7 \text{ cm.}$$

$$\text{Answer } AP = \dots\dots\dots 24 \dots\dots\dots \text{cm, } PB = \dots\dots\dots 7 \dots\dots\dots \text{cm [3]}$$





21 (a)



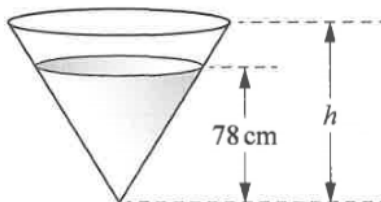
The diagram shows a cylindrical container.
It is 65% full of water.
The depth of the water is 78 cm.

Calculate the height of the container.

$$\begin{aligned}\text{height of container} &= \frac{100}{65} \times 78 \\ &= 120 \text{ cm.}\end{aligned}$$

Answer 120 cm [2]

(b)



The diagram shows a conical container.
It is 65% full of water.
The depth of the water is 78 cm.

Calculate the height, h , of the container.

$$\begin{aligned}\left(\frac{78}{h}\right)^3 &= \frac{65}{100} \\ \frac{78}{h} &= \sqrt[3]{\frac{65}{100}} \\ h &= 78 \div \sqrt[3]{\frac{65}{100}} \\ &= 90.0\end{aligned}$$

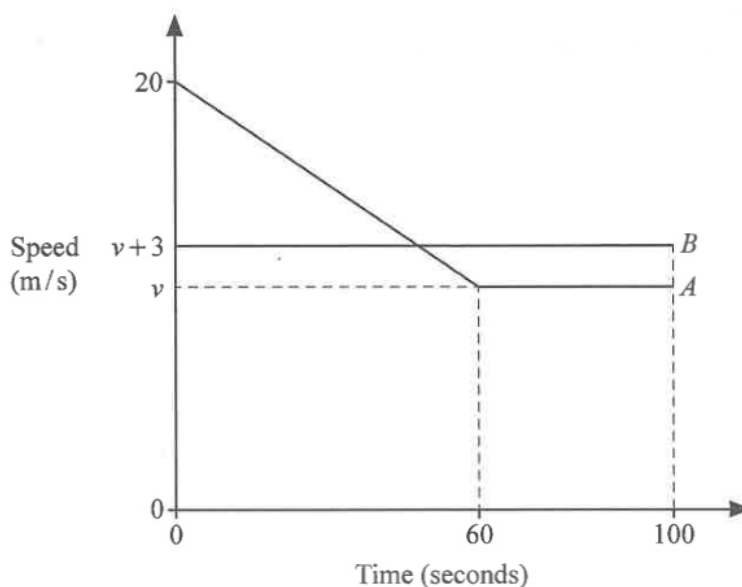
Answer $h =$ 90.0 cm [3]

0022221558315



15

- 22 The diagram shows speed–time graphs for two cyclists, A and B , travelling along the same road in the same direction.



A decelerates from 20 m/s to $v\text{ m/s}$ and then travels at a constant speed of $v\text{ m/s}$.

B travels at a constant speed of $(v+3)\text{ m/s}$.

After 100 seconds B has travelled 75 m further than A .

The area beneath each speed–time graph represents the distance travelled by each cyclist.

Find the value of v .

$$\begin{aligned}\text{Distance travelled by } A &= \frac{1}{2}(20+v)(60) + v(100-60) \\ &= 30(20+v) + 40v \\ &= (600 + 70v)\text{ m}.\end{aligned}$$

$$\text{Distance travelled by } B = 100(v+3) = 100v + 300$$

$$100v + 300 = 600 + 70v + 75$$

$$30v = 375$$

$$v = 12.5$$

Answer $v = 12.5$ [4]



* 0022221558316 *



16

23 Solve the equation $\frac{2x}{x+4} + \frac{5}{3x-5} = 3$.

Show all your working.

Give your solutions correct to 2 decimal places.

$$2x(3x-5) + 5(x+4) = 3(x+4)(3x-5)$$

$$6x^2 - 10x + 5x + 20 = 3(3x^2 + 12x - 5x - 20)$$

$$6x^2 - 5x + 20 = 3(3x^2 + 7x - 20)$$

$$6x^2 - 5x + 20 = 9x^2 + 21x - 60$$

$$0 = 3x^2 + 26x - 80$$

$$x = \frac{-26 \pm \sqrt{26^2 - 4(3)(-80)}}{2(3)}$$

$$= \frac{-26 \pm \sqrt{1636}}{6}$$

$$x = 2.41 \text{ or } -11.07$$

Answer $x = 2.41$ or $x = -11.07$ [5]

24 Simplify $\left(\frac{16}{81a^4c^{12}}\right)^{-\frac{3}{4}}$, giving your answer in the form pa^qc^r , where p , q and r are constants.

$$= \left(\frac{81a^4c^{12}}{16}\right)^{\frac{3}{4}}$$

$$= \frac{81^{\frac{3}{4}}a^3c^9}{16^{\frac{3}{4}}}$$

$$= \frac{81 \cdot a^3 c^9}{16^{\frac{3}{4}}} = \frac{27a^3c^9}{8} = \frac{27}{8}a^3c^9$$

Answer $\frac{27}{8}a^3c^9$ [2]



* 0022221558317 *



17

25 The table shows information about the masses of some kiwi fruits.

Mass (m grams)	$70 < m \leq 75$	$75 < m \leq 77$	$77 < m \leq 79$	$79 < m \leq 81$	$81 < m \leq 85$
Frequency	8	8	7	5	4

(a) Calculate an estimate for

(i) the mean mass,

Answer 77.0625 g [1]

(ii) the standard deviation of the masses.

Answer 3.38 g [1]

(b) The masses of some plums have a mean of 65.5 grams and a standard deviation of 3.83 grams.

Make two comments comparing the (1.) averages and (2.) distributions of the masses of the kiwi fruits and the plums.

- Plums weigh less on average as mean mass of plums is lower at 65.5g while for kiwi fruits is at 77.0625g.
- Spread of masses of plums is wider as standard deviation of masses of plums is greater at 3.83g compared to that of kiwi fruits at 3.38g.

[2]

mass of plum is greater at 3.83g compared to that of kiwi fruits at 3.38g.

[2]

© UCLES & MOE 2025



4052/01/O/N/25

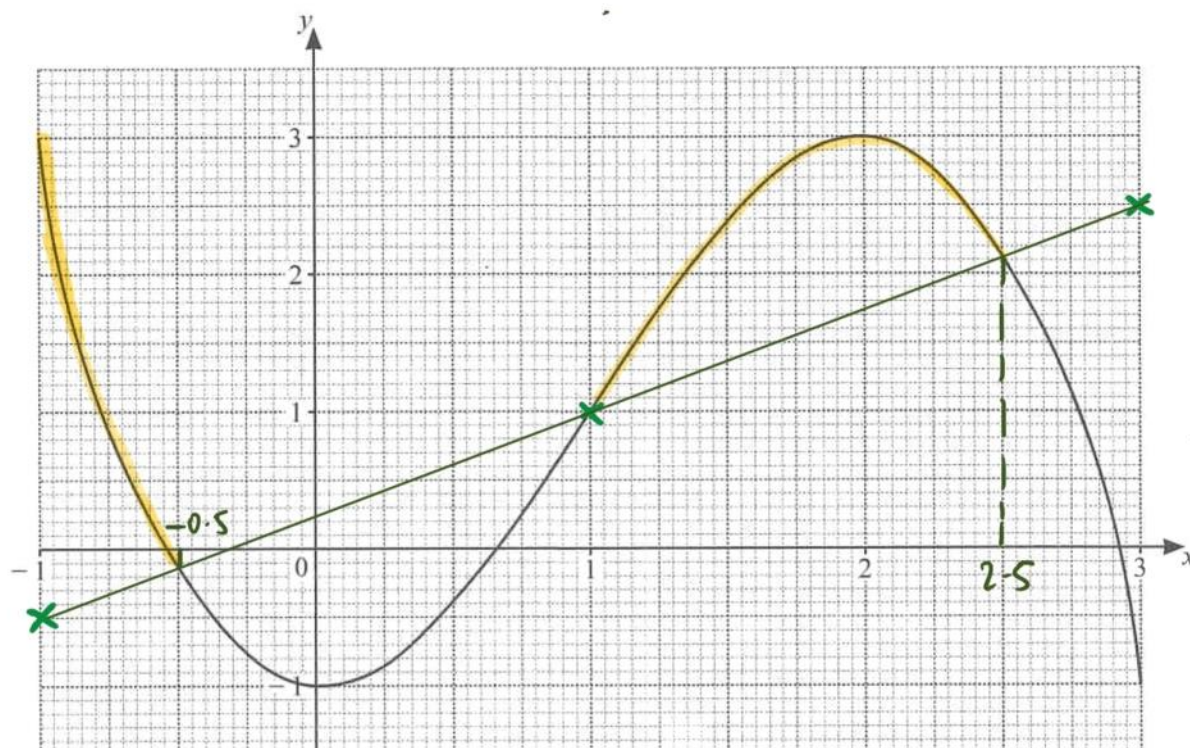
[Turn over]

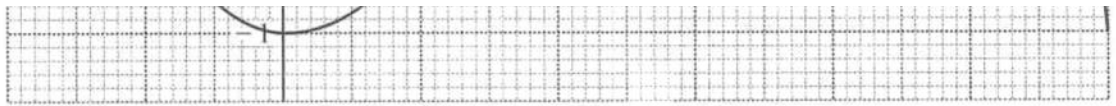
* 0022221558318 *



18

26 The graph of $y = 3x^2 - x^3 - 1$ is drawn on the grid.





By drawing a suitable straight line on the grid, solve the inequality $4x^3 - 12x^2 + 3x + 5 < 0$.

$$4x^3 - 12x^2 + 3x + 5 < 0$$

$$\text{Draw } y = \frac{3}{4}x + \frac{1}{4}$$

$$\div -4: -x^3 + 3x^2 - \frac{3}{4}x - \frac{5}{4} > 0$$

$$-x^3 + 3x^2 > \frac{3}{4}x + \frac{5}{4}$$

$$3x^2 - x^3 - 1 > \frac{3}{4}x + \frac{1}{4}$$

x	-1	1	3
y	-0.5	1	2.5

Answer $x < -0.5$ or $1 < x < 2.5$ [4]



* 0022221558319 *



19

- 27 Abid has two fair numbered spinners, Spinner X and Spinner Y .
The probability that Spinner X lands on an odd number is x .

- (a) Write down an expression for the probability that Spinner X lands on an even number.

Answer $1 - x$ [1]

- (b) Abid spins the two spinners.
The probability that both spinners land on an odd number is $\frac{3}{20}$.
The probability that Spinner X lands on an even number and Spinner Y lands on an odd number is $\frac{9}{20}$.

Find the probability that both spinners land on an even number.

$$\text{Let } P(\text{spinner } Y \text{ on odd}) = y.$$

spinner X y / odd spinner Y

$$xy = \frac{3}{20}$$

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

© UCLES & MOE 2025



4052/01/O/N/25